

Physical Chemistry (I)

Lecture 13

Chemical Potential



In the Last Class

- Helmholtz and Gibbs energies
 - Definitions
 - Meanings
- Thermodynamics potentials
 - Differential forms
 - Properties of exact differentials
- Thermochemistry of Gibbs energy



Last class

Helmholtz and Gibbs Energies

$$\text{Helmholtz energy } A = U - TS$$

$$\text{Gibbs energy } G = H - TS$$

Criteria of spontaneity:

$$dA_{T,V} \leq 0$$

(At constant T and V , $dA \propto -dS_{\text{total}}$)

$$dG_{T,p} \leq 0$$

(At constant T and p , $dG \propto -dS_{\text{total}}$)



Last class

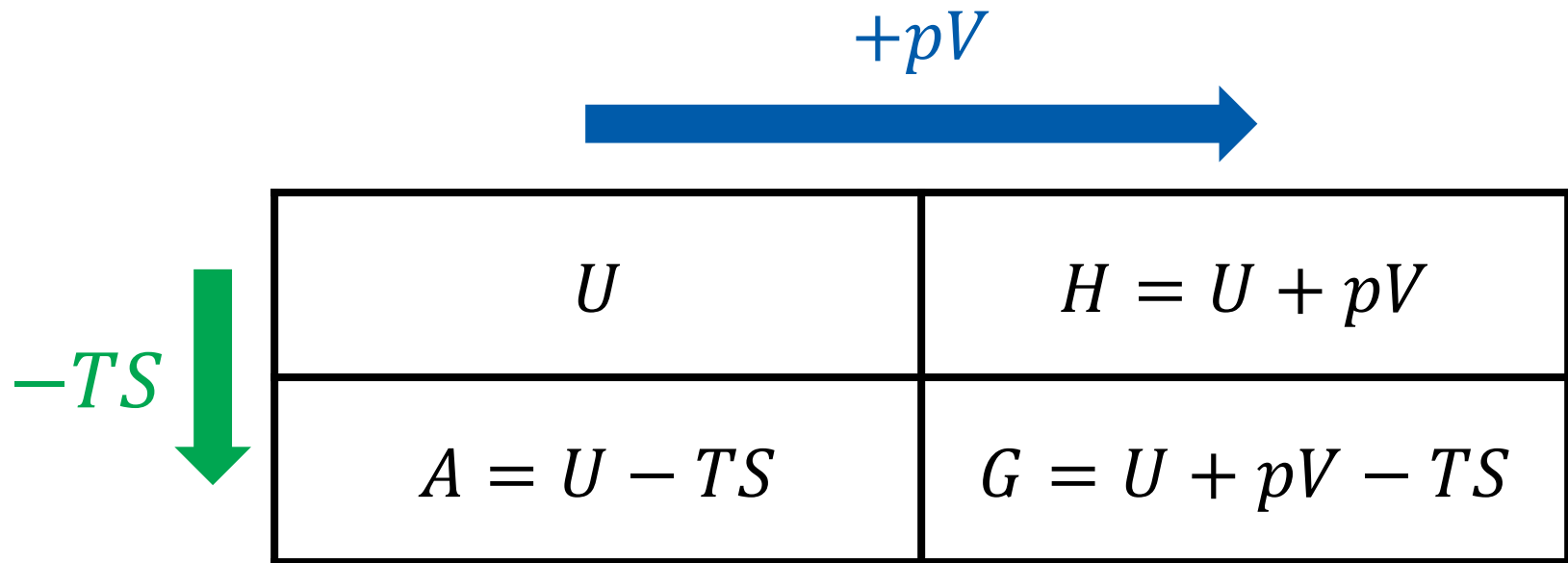
So, What Do They Mean?

- Helmholtz energy = maximum work
- Gibbs energy = maximum non-expansion work



Last class

Thermodynamic Potentials



Last class

Natural Independent Variables

- Internal energy $U = U(S, V)$

$$dU = TdS - pdV$$

- Enthalpy $H = U + pV = H(S, p)$

$$dH = TdS + Vdp$$

- Helmholtz energy $A = U - TS = A(T, V)$

$$dA = -SdT - pdV$$

- Gibbs energy $G = U + pV - TS = G(T, p)$

$$dG = -SdT + Vdp$$



Last class

Gibbs-Helmholtz Equation

$$\left(\frac{\partial(G/T)}{\partial T} \right)_p = -\frac{H}{T^2}$$

At constant T ,

$$\left(\frac{\partial(\Delta G/T)}{\partial T} \right)_p = -\frac{\Delta H}{T^2}$$



Maxwell Relations

For an exact differential $df = gdx + hdy$,

$$\left(\frac{\partial g}{\partial y}\right)_x = \left(\frac{\partial h}{\partial x}\right)_y$$

- Internal energy $dU = TdS - pdV$

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$$

- Enthalpy $dH = TdS + Vdp$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$$

Maxwell Relations

For an exact differential $df = gdx + hdy$,

$$\left(\frac{\partial g}{\partial y}\right)_x = \left(\frac{\partial h}{\partial x}\right)_y$$

- Helmholtz energy $dA = -SdT - pdV$

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$$

- Gibbs energy $dG = -SdT + Vdp$

$$\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p$$

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Thermodynamic Eq. of State

$$p = T \left(\frac{\partial p}{\partial T} \right)_V - \pi_T$$

Thermochemistry of G

$$G = H - TS$$

At constant T ,

$$\Delta G = \Delta H - T\Delta S$$

We can define the standard Gibbs energy of reaction:

$$\Delta_r G^\ominus = \Delta_r H^\ominus - T\Delta_r S^\ominus$$

and the standard Gibbs energy of formation:

$$\Delta_f G^\ominus = \Delta_f H^\ominus - T\Delta_f S^\ominus$$

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Pressure Dependence of G

$$\Delta G_m = \int_{p_i}^{p_f} V_m dp$$

1. Condensed phase: $V_m \approx \text{constant}$
2. Gas phase: $V_m = V/n = RT/p$

6. Phase Transitions

Chapter Overview

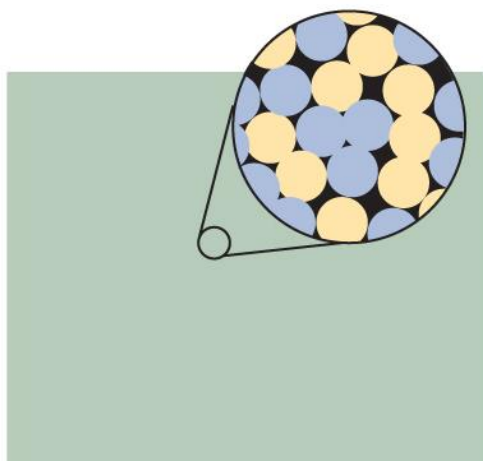
- Phase and chemical potential
 - Phase and phase transition
 - Chemical potential
 - Phase diagram and phase rule
- van der Waals fluid
 - Liquid-gas phase transition
 - Supercritical fluid
 - Critical point and the principle of corresponding states
- Thermodynamic aspects of phase transitions



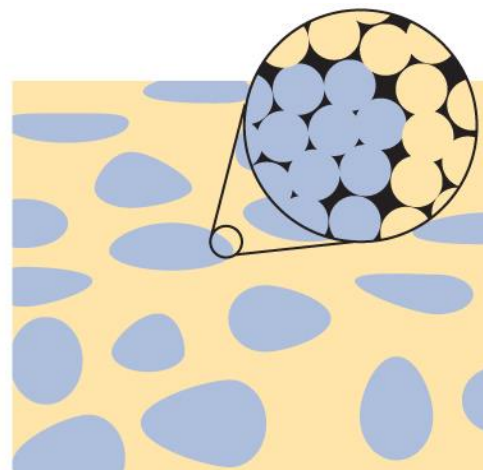
Phase

A form of matter that is **uniform** throughout
in **chemical composition** and **physical state**

Solution: Single Phase



solution
(one phase)



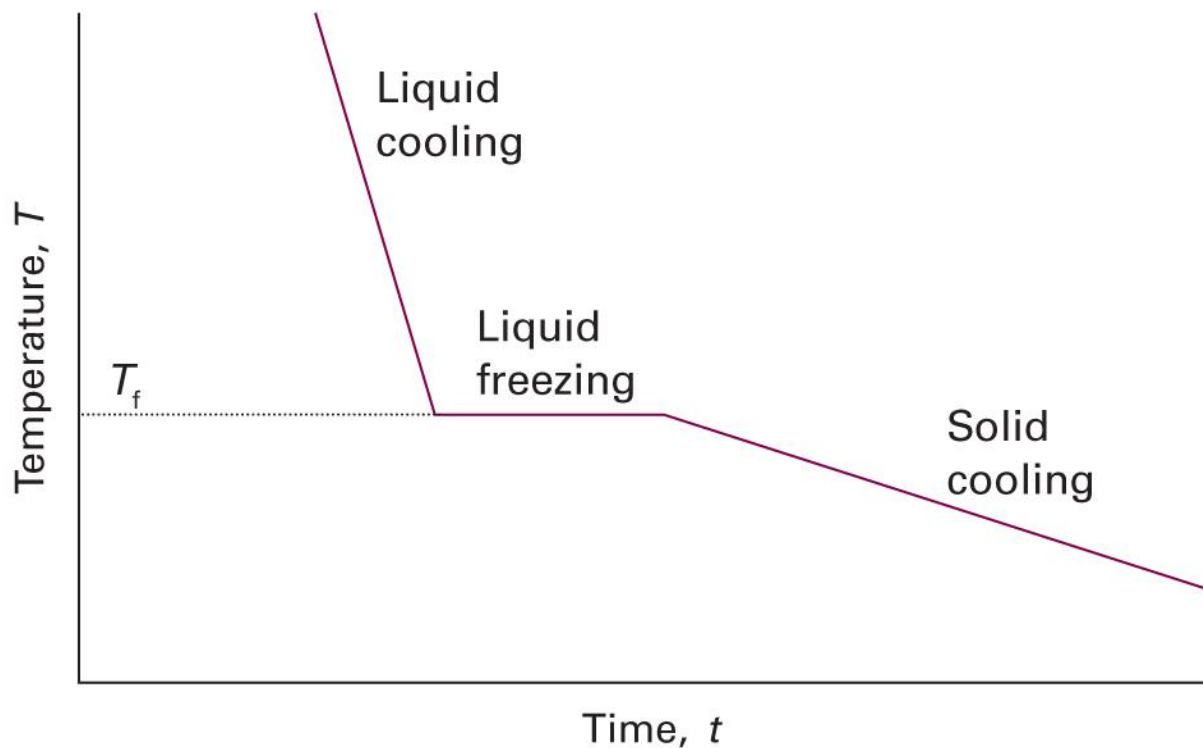
dispersion
(two phases)

Phase Transition

Spontaneous conversion
of one phase into another phase

- occurs at transition temperature T_{trs} (for given p)
- At T_{trs} , the two phases are at equilibrium

Thermal Analysis



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Two Phases at Equilibrium

$$dU = TdS - pdV$$

- Mechanistic equilibrium: $p_1 = p_2$
 - Conjugate extensive variable: dV
- Thermal equilibrium: $T_1 = T_2$
 - Conjugate extensive variable: $dS = \delta q/T$
- Diffusive equilibrium: ?
 - Conjugate extensive variable: dn



Chemical Potential

- Diffusive equilibrium: $\mu_1 = \mu_2$
 - Conjugate extensive variable: dn
- (Modified) differentials of thermodynamic potentials

$$dU = TdS - pdV + \mu dn$$

$$dH = TdS + Vdp + \mu dn$$

$$dA = -SdT - pdV + \mu dn$$

$$dG = -SdT + Vdp + \mu dn$$

3 Independent Variables

$$dU = TdS - pdV + \mu dn$$

This implies that $U = U(S, V, n)$, which leads to

$$dU = \left(\frac{\partial U}{\partial S}\right)_{V,n} dS + \left(\frac{\partial U}{\partial V}\right)_{S,n} dV + \left(\frac{\partial U}{\partial n}\right)_{S,V} dn$$

Comparison gives

$$\left(\frac{\partial U}{\partial S}\right)_{V,n} = T, \quad \left(\frac{\partial U}{\partial V}\right)_{S,n} = -p, \quad \left(\frac{\partial U}{\partial n}\right)_{S,V} = \mu$$

Meaning of μ

$$dG = -SdT + Vdp + \mu dn$$

This implies that $G = G(T, p, n)$.

G is extensive, and n is the only extensive variable. Thus,

$$G(T, p, n) = ng(T, p)$$

and

$$\left(\frac{\partial G}{\partial n}\right)_{T,p} = \left[\frac{\partial\{ng(T, p)\}}{\partial n}\right]_{T,p} = g(T, p)$$

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Meaning of μ

$$g(T, p) = \frac{G}{n} = \left(\frac{\partial G}{\partial n} \right)_{T, p}$$

However,

$$dG = -SdT + Vdp + \mu dn$$

gives

$$\mu = \left(\frac{\partial G}{\partial n} \right)_{T, p} = g(T, p) = \frac{G}{n}$$

(molar Gibbs energy)

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Thermodynamic Potentials

$$\mu = G/n$$

can be written as

$$G = \mu n$$

From this equation, we can obtain

$$H = G + TS = \mu n + TS$$

$$A = G - pV = \mu n - pV$$

$$U = G + TS - pV = \mu n + TS - pV$$



Gibbs-Duhem Equation

$$U = \mu n + TS - pV$$

Differentiating both sides,

$$dU = \mu dn + n d\mu + T dS + S dT - p dV - V dp$$

but we know that

$$dU = T dS - p dV + \mu dn$$

Thus,

$$S dT - V dp + n d\mu = 0 \text{ (Gibbs-Duhem equation)}$$

$$d\mu = -S_m dT + V_m dp$$



Two Systems in Contact

$$dG = -SdT + Vdp + \mu dn$$

At constant T and p ,

$$dG = \mu dn$$

For two systems in contact,

$$dG_1 = \mu_1 dn_1$$

$$dG_2 = \mu_2 dn_2$$

The total change in Gibbs energy is

$$dG = dG_1 + dG_2 = \mu_1 dn_1 + \mu_2 dn_2$$

Diffusive Equilibrium

$$dG = dG_1 + dG_2 = \mu_1 dn_1 + \mu_2 dn_2$$

If the two subsystems consist of the same type of particles and can exchange particles,

$$dn_1 + dn_2 = 0$$

Hence,

$$dG = \mu_1 dn_1 - \mu_2 dn_1 = (\mu_1 - \mu_2) dn_1$$

At equilibrium, $dG = 0$, and

$$\mu_1 = \mu_2$$

What If $\mu_1 \neq \mu_2$?

$$dG = \mu_1 dn_1 - \mu_2 dn_1 = (\mu_1 - \mu_2) dn_1$$

- If $\mu_1 > \mu_2$, for $dG < 0$ (spontaneous change),

$$dn_1 < 0 \text{ and } dn_2 = -dn_1 > 0,$$

which means that the particles move from system 1 to 2

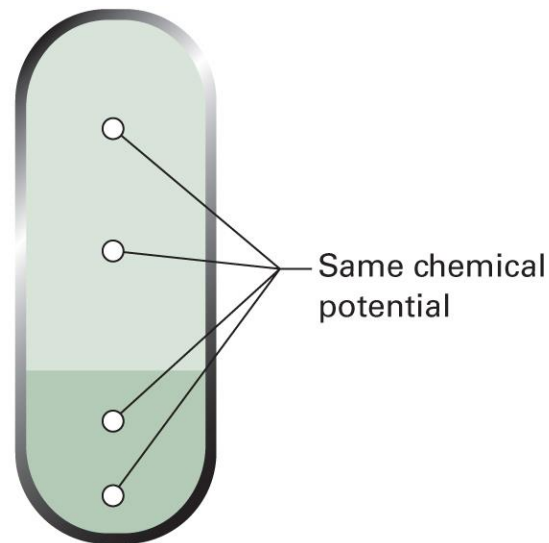
- If $\mu_1 < \mu_2$, for $dG < 0$ (spontaneous change),

$$dn_1 > 0 \text{ and } dn_2 = -dn_1 < 0,$$

which means that the particles move from system 2 to 1

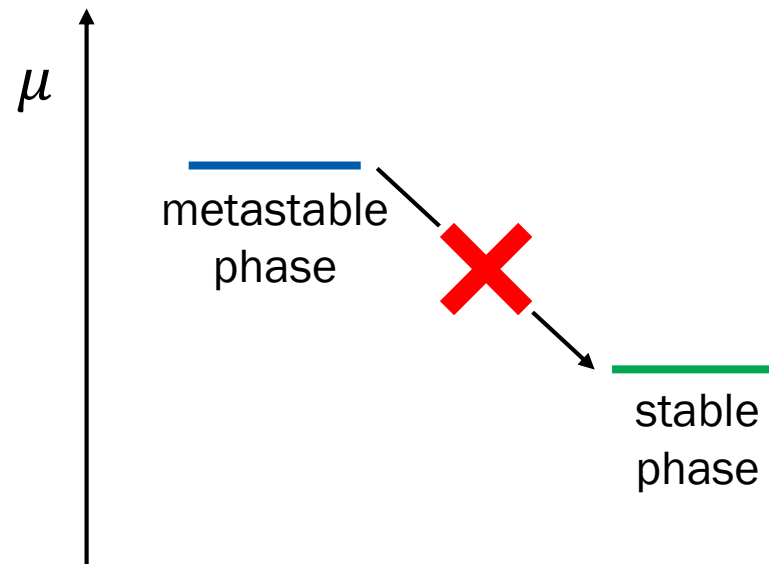
Criteria for Phase Stability

At equilibrium, the **chemical potential** of a substance is the **same** in and throughout every phase in the system



Metastable Phase

Thermodynamically unstable phases that persist because the transition is kinetically hindered

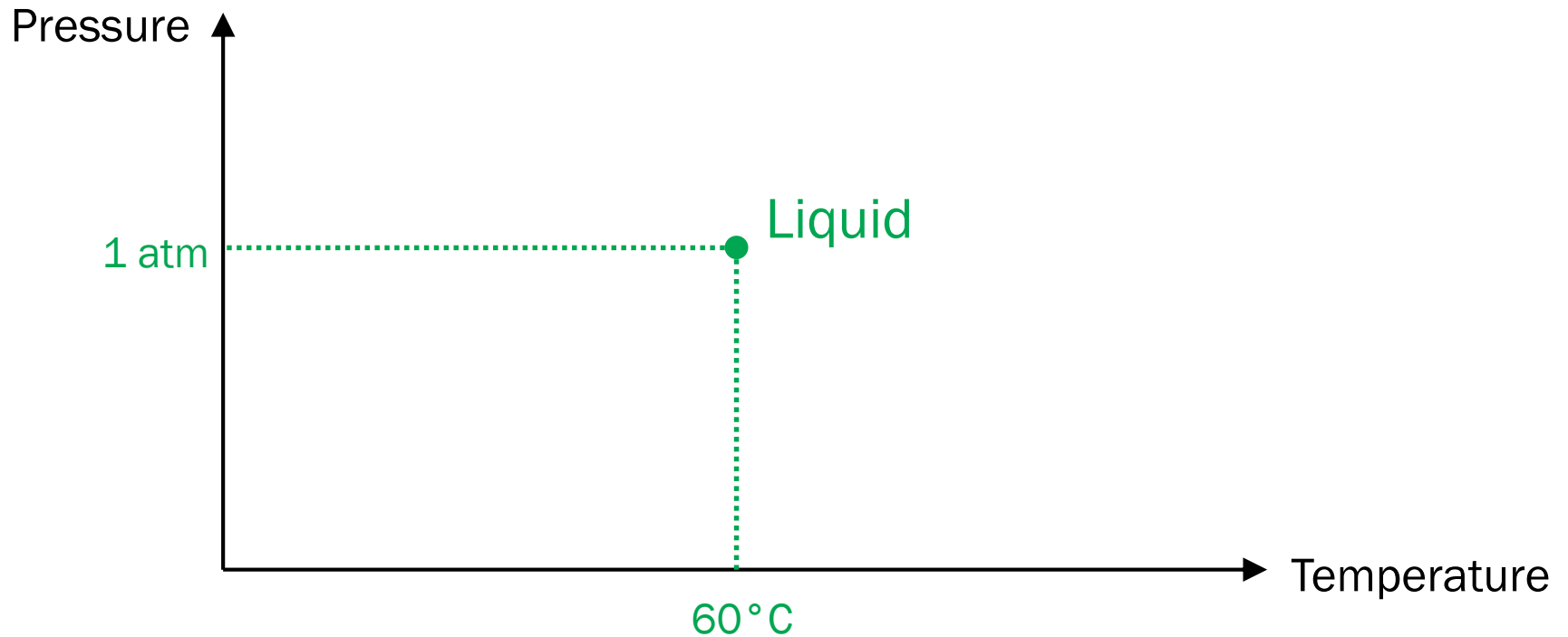


Phase Diagram

A diagram that shows the regions at which its various phases are **thermodynamically stable**

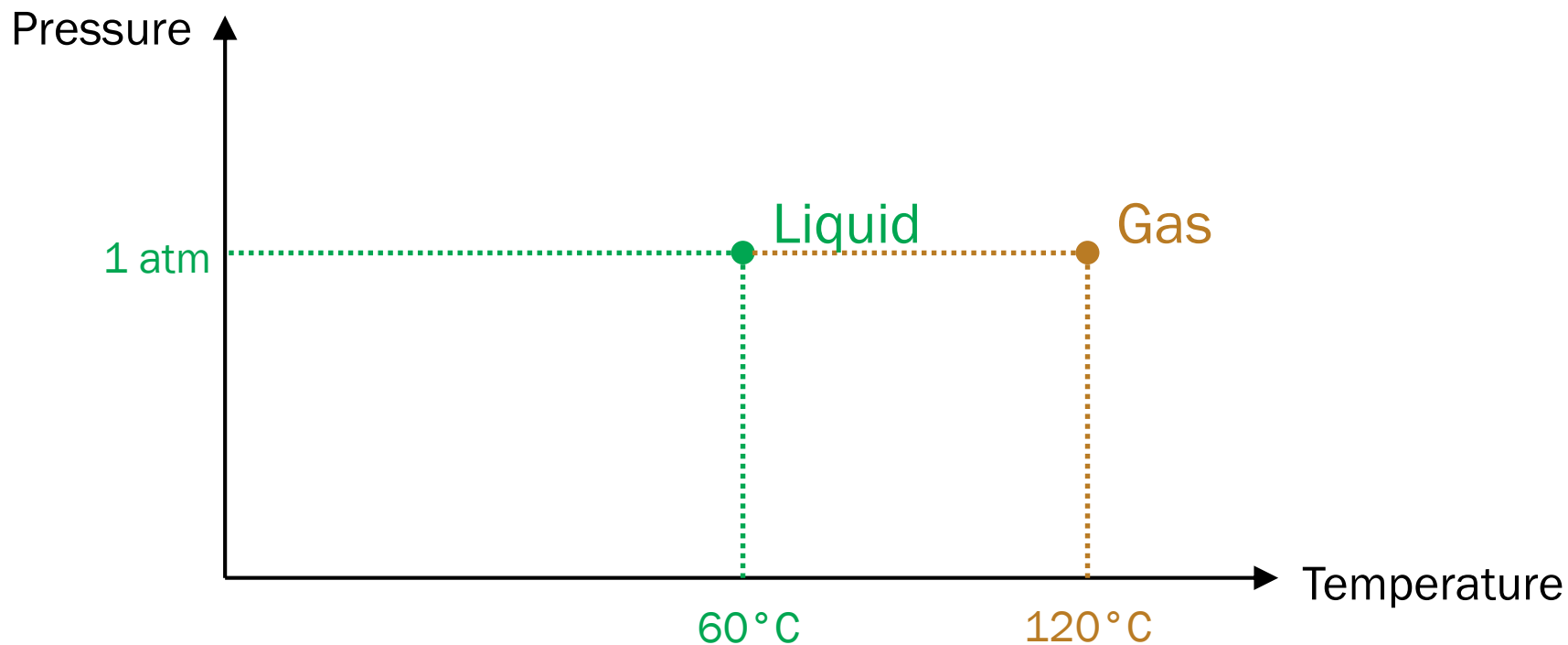
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Drawing Phase Diagram



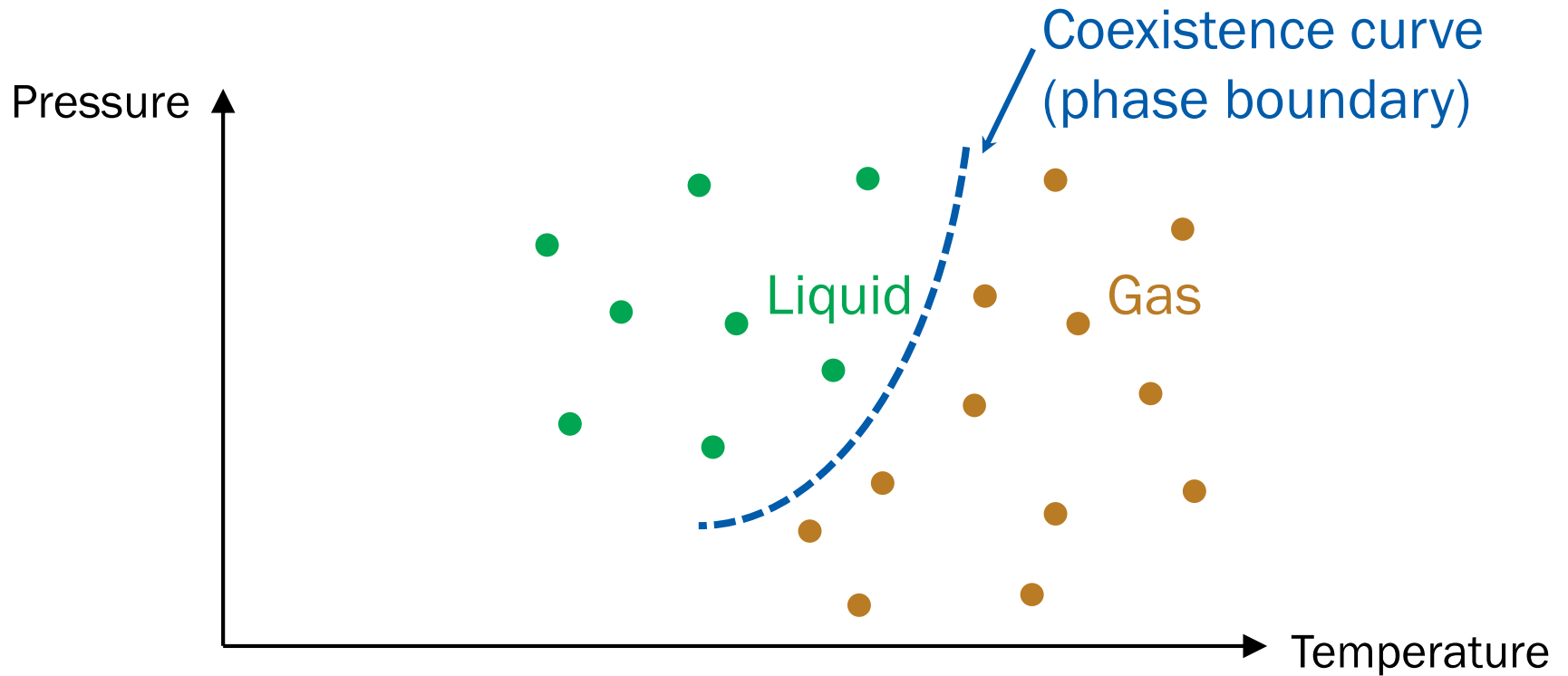
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Drawing Phase Diagram

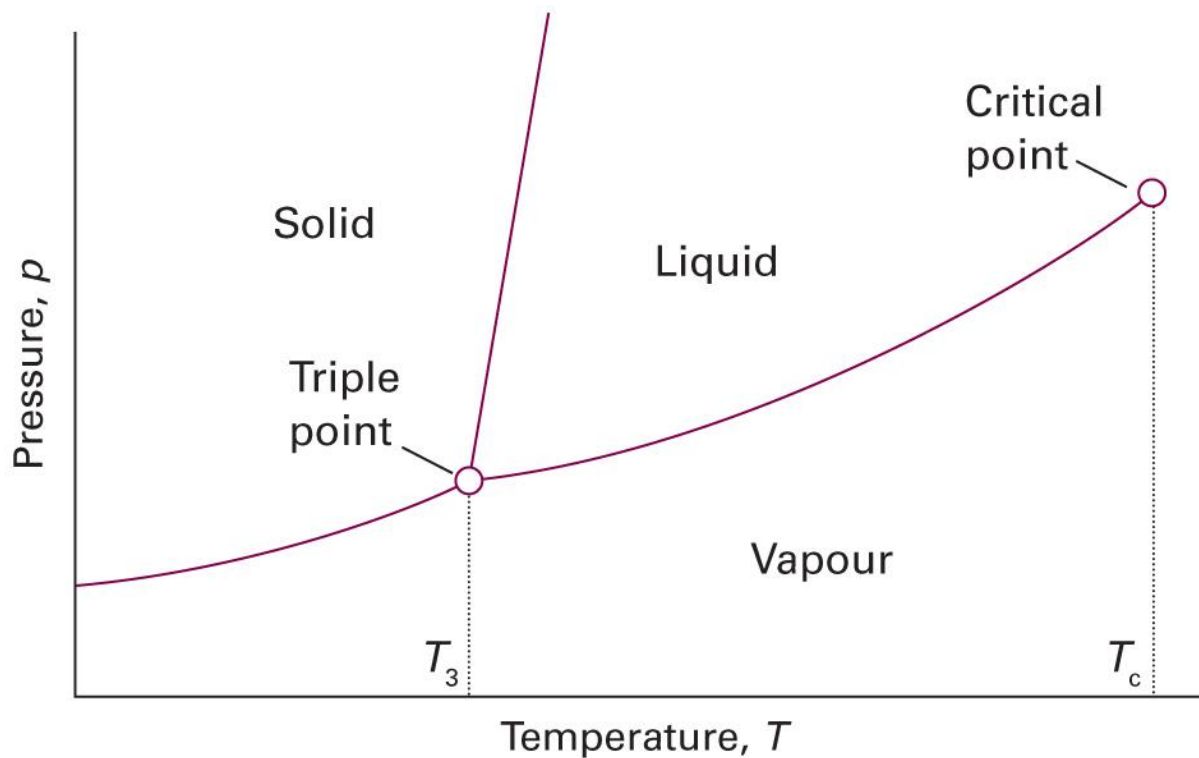


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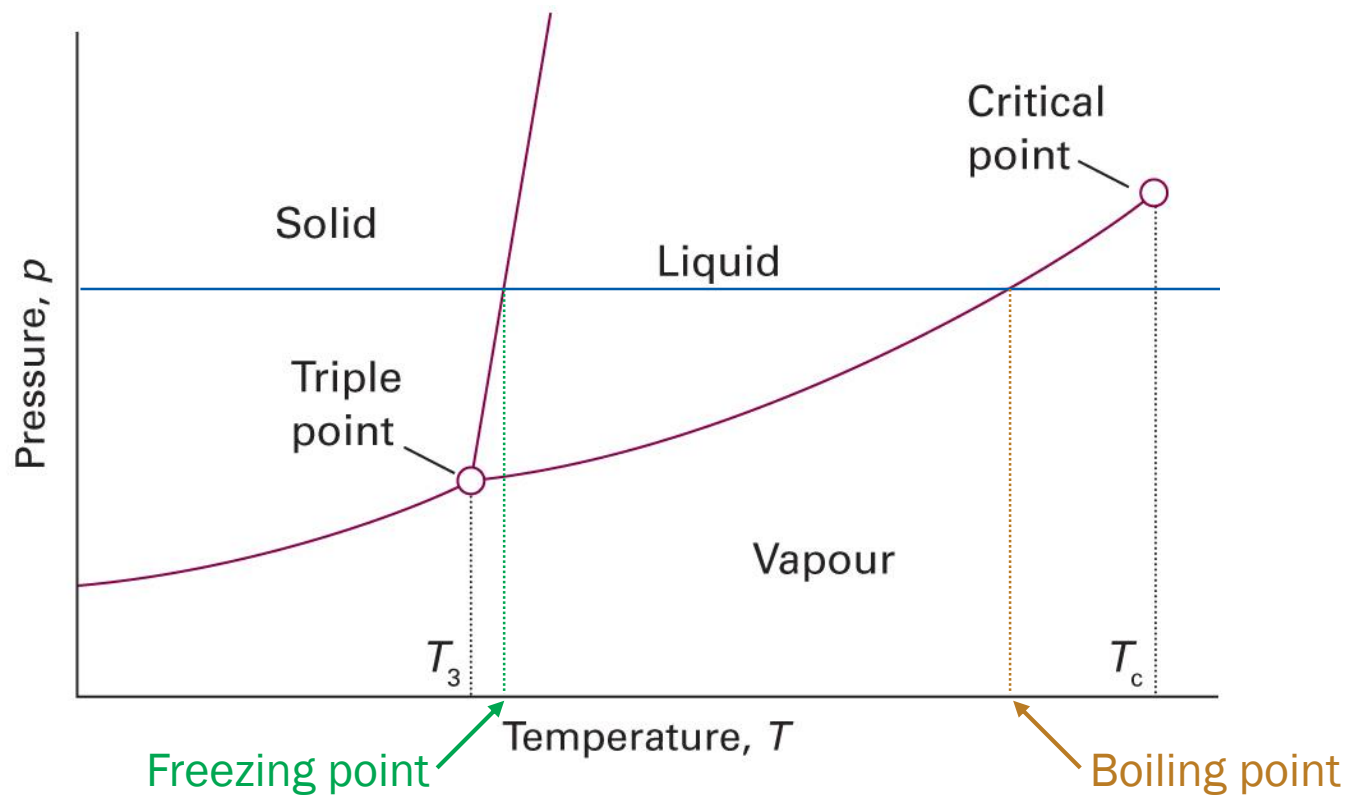
Drawing Phase Diagram



Typical Phase Diagram



Freezing and Boiling Points



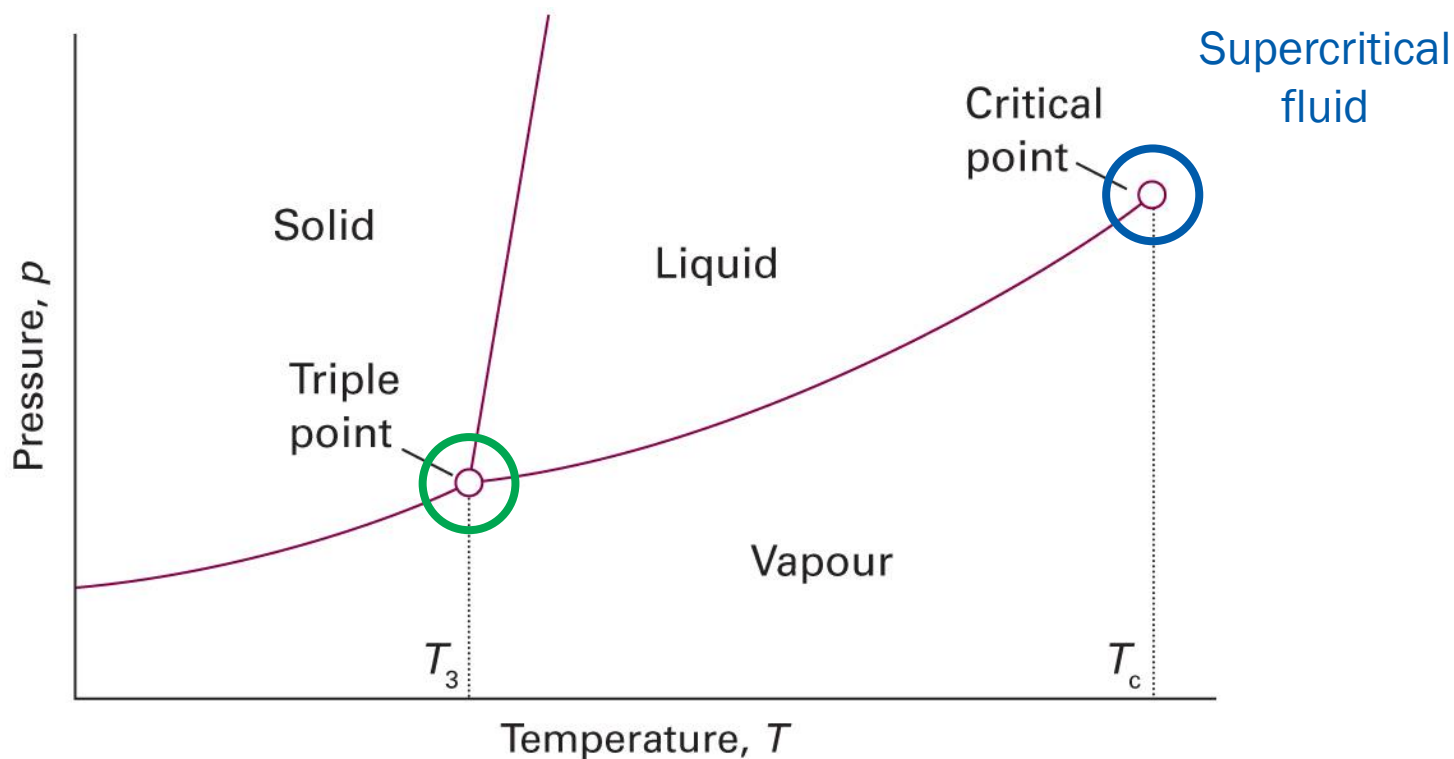
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.4)

Freezing and Boiling Points

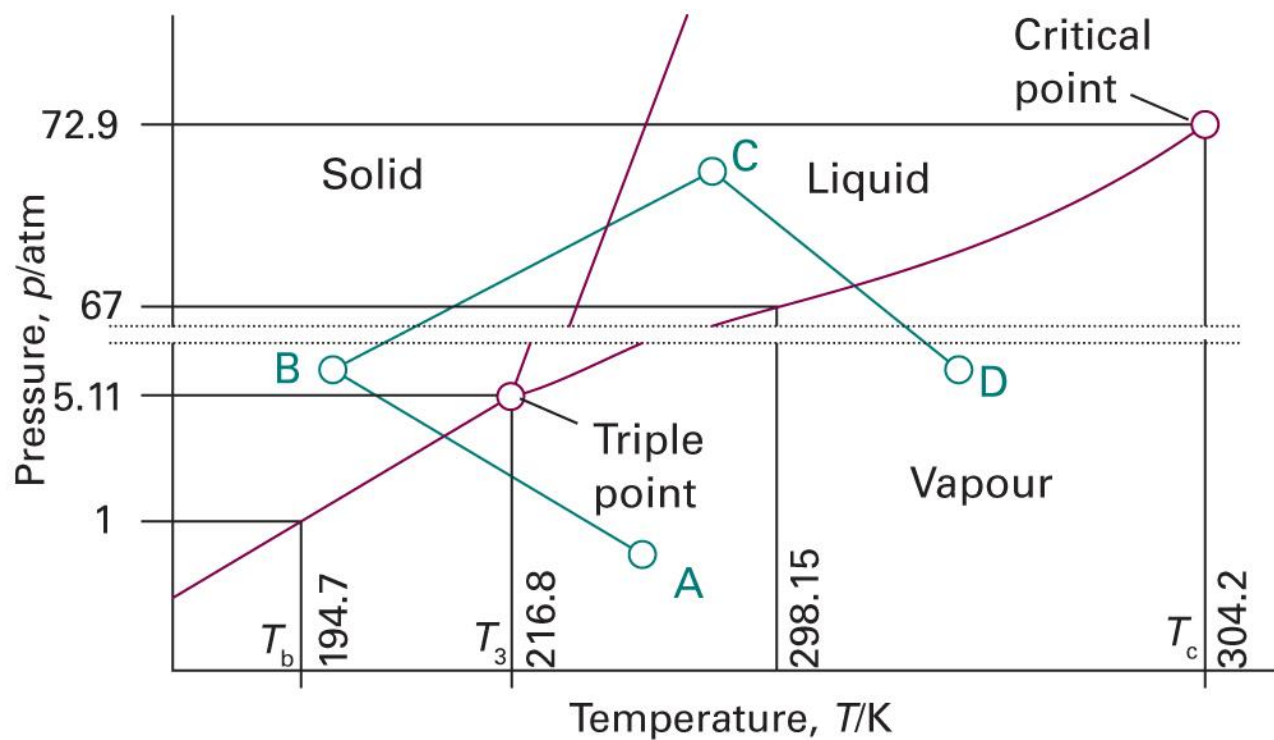
- Normal boiling point T_b : b.p. at 1 atm
- Standard boiling point: b.p. at 1 bar
- e.g. water: $T_b = 100^\circ\text{C}$, standard b.p. = 99.6°C

- Normal freezing point T_f : f.p. at 1 atm
- Standard freezing point: f.p. at 1 bar
- Usually the difference between the two is negligible

Two Characteristic Points

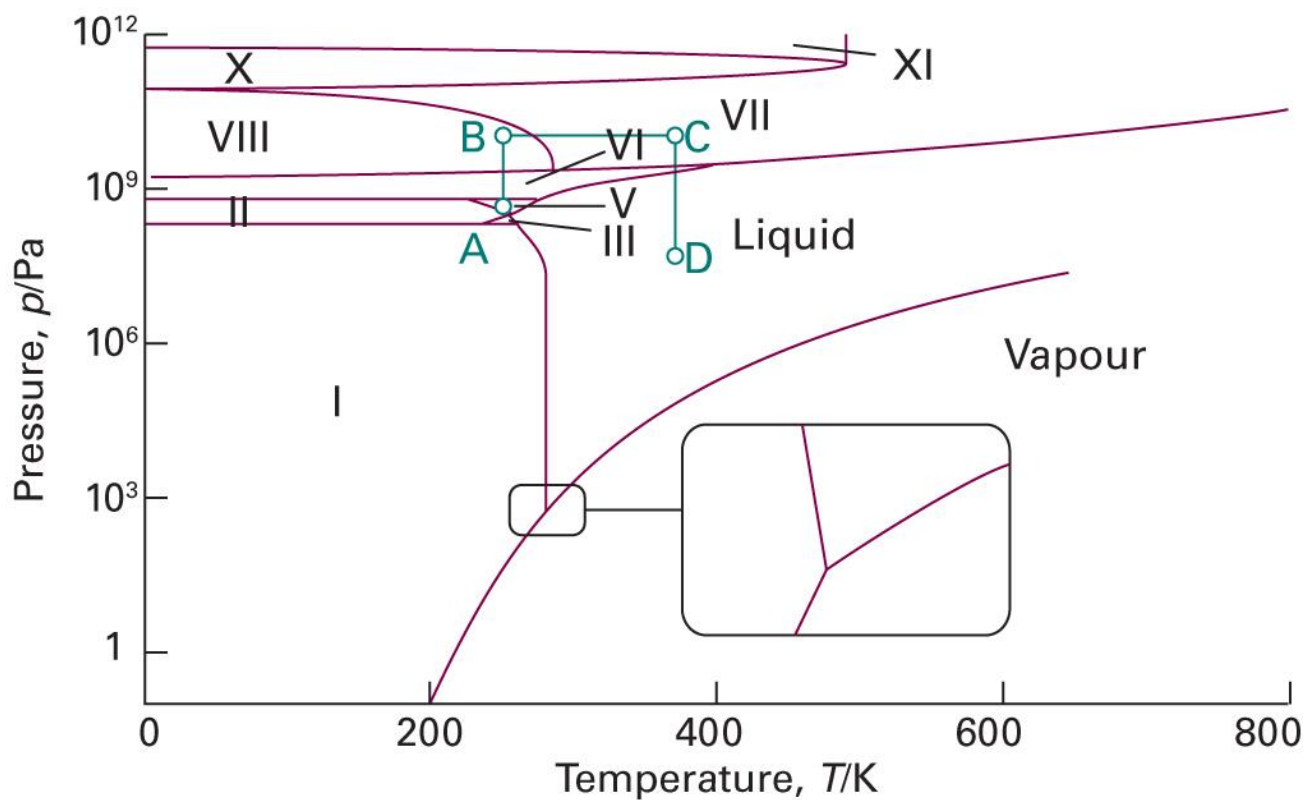


Carbon Dioxide



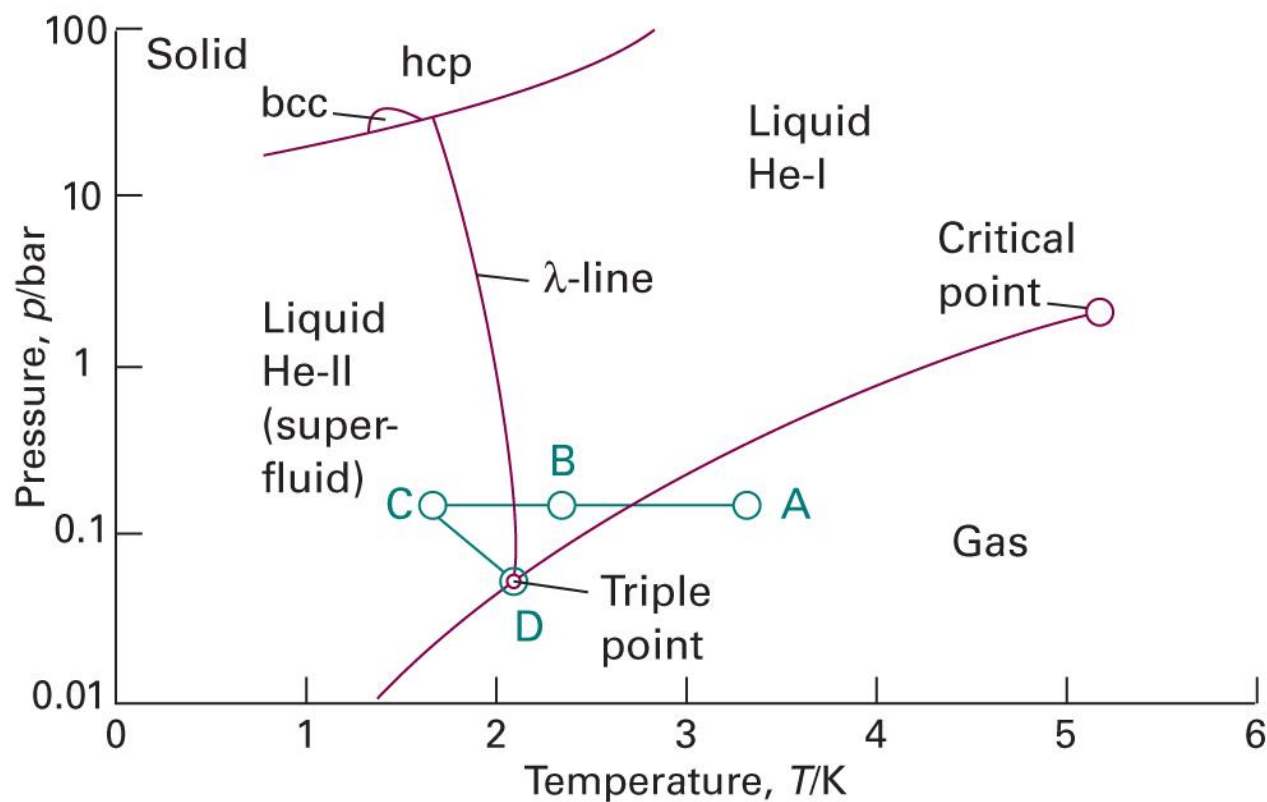
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.8)

Water



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.9)

Helium (^4He)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.11)

Phase Rule

“Can we have a quadruple point?”

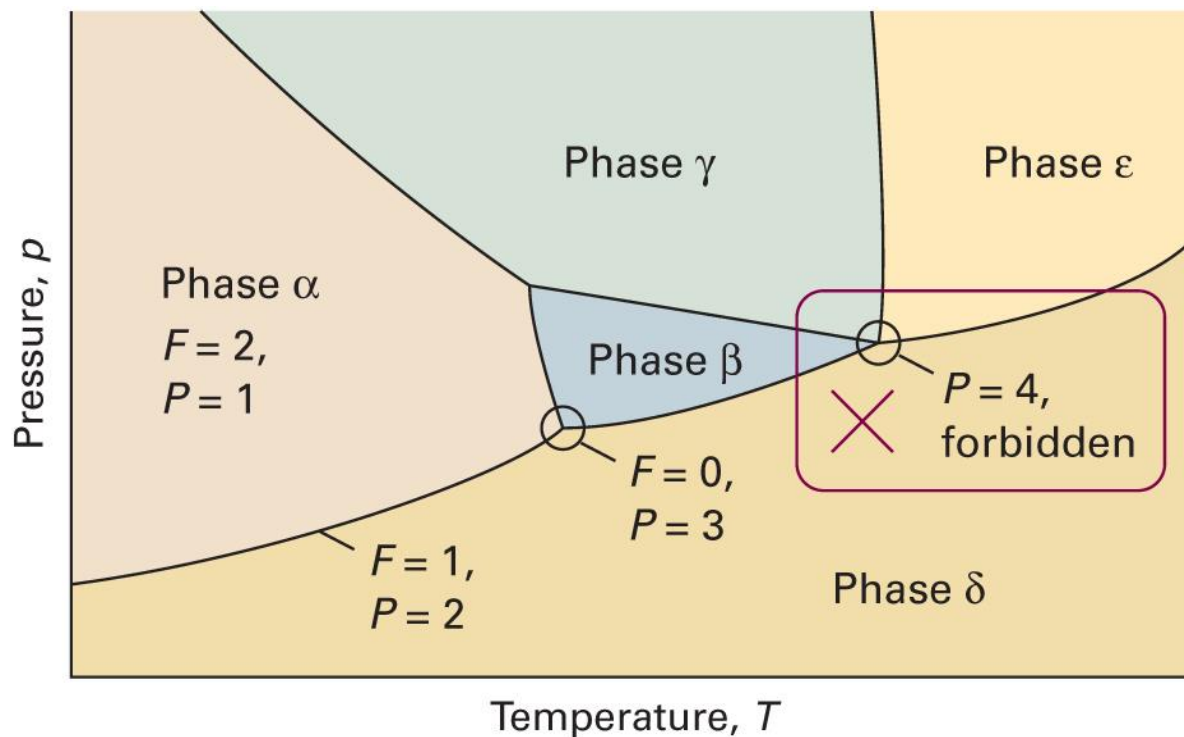
- **Phase rule:** a general relation between
 - Number of degrees of freedom F
 - Number of components C
 - Number of phases at equilibrium P
- Use the **criteria for phase stability** (same μ)
 - Recall that $\mu = g(T, p)$

Single Component ($C = 1$)

- $P = 1$: both T and p can be independently varied
 $\therefore F = 2$ (bivariant)
- $P = 2$: $\mu_1(T, p) = \mu_2(T, p)$
 $\therefore F = 1$ (univariant)
- $P = 3$: $\mu_1(T, p) = \mu_2(T, p)$ and $\mu_2(T, p) = \mu_3(T, p)$
 $\therefore F = 0$ (invariant)
- $P = 4$: two variables and three restrictions $\rightarrow$ impossible

In summary, $F = 3 - P$

Single Component ($C = 1$)



Multiple Components ($C > 1$)

- The total number of independent intensive variables
 - T and p : 2
 - $(C - 1)$ mole fractions (-1 due to normalization) for each phase: $P(C - 1)$
 - Total number = $P(C - 1) + 2$
- The total number of equations (restrictions)
 - $\mu_J(\text{phase 1}; p, T) = \mu_J(\text{phase 2}; p, T) = \dots$ for component J
 - Total number = $C(P - 1)$
- Therefore, $F = P(C - 1) + 2 - C(P - 1) = C - P + 2$

